

```
akshi@akshita:~$ mkdir filecopy_lab
mkdir: filecopy_lab: File exists
akshi@akshita:~$ cd filecopy_lab
akshi@akshita:~/filecopy_lab$ pwd
/home/akshi/filecopy_lab
akshi@akshita:~/filecopy_lab$ nano sample.txt
akshi@akshita:~/filecopy_lab$ gcc filecopy.c -o filecopy
cc1: fatal error: filecopy.c: No such file or directory
compilation terminated.
akshi@akshita:~/filecopy_lab$ ls
sample.txt
akshi@akshita:~/filecopy_lab$ pwd
/home/akshi/filecopy_lab
akshi@akshita:~/filecopy_lab$ ls
sample.txt
akshi@akshita:~/filecopy_lab$ nano filecopy.c
akshi@akshita:~/filecopy_lab$
```

```
akshi@akshita:~/filecopy_lab$  
akshi@akshita:~/filecopy_lab$ #include <stdio.h>  
#include <stdlib.h>  
#include <fcntl.h>  
#include <unistd.h>  
  
#define BUFFER_SIZE 4096  
  
int main(int argc, char *argv[])  
{  
    int src, dest;  
    char buffer[BUFFER_SIZE];  
    ssize_t bytes_read, bytes_written;  
  
    if (argc != 3) {  
        printf("Usage: %s <source> <destination>\n", argv[0]);  
        return 1;  
    }  
  
    src = open(argv[1], O_RDONLY);  
  
    if (src == -1) {  
        perror("Error opening source file");  
        return 1;  
    }  
  
    dest = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);  
  
    if (dest == -1) {  
        perror("Error opening destination file");  
        close(src);  
        return 1;  
    }  
}
```

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    }

    if (bytes_read == -1) {
        perror("Error reading file");
    }

    close(src);
    close(dest);

    return 0;
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

#define BUFFER_SIZE 4096

int main(int argc, char *argv[])

    int src, dest;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    if (argc != 3) {
kshi@akshita:~/filecopy_lab$ #include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

#define BUFFER_SIZE 4096

int main(int argc, char *argv[])

    int src, dest;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    if (argc != 3) {
        printf("Usage: %s <source> <destination>\n", argv[0]);
        return 1;
    }

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        return 1;
    }

    while ((bytes_read = read(src, buffer, BUFFER_SIZE)) > 0) {

        bytes_written = write(dest, buffer, bytes_read);

        if (bytes_written != bytes_read) {
            perror("Error writing file");
            close(src);
            close(dest)
akshi@akshita:~/filecopy_lab$ #include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

#define BUFFER_SIZE 4096

int main(int argc, char *argv[])
{
    int src, dest;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    if (argc != 3) {
        printf("Usage: %s <source> <destination>\n", argv[0]);
        return 1;
    }

    src = open(argv[1], O_RDONLY);

    if (src == -1) {
        perror("Error opening source file");
        return 1;
    }

    dest = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (dest == -1) {
        perror("Error opening destination file");
        close(src);
        return 1;
    }
}

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```
        close(src);
        close(dest);
akshi@akshita:~/filecopy_lab$ #include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

#define BUFFER_SIZE 4096

int main(int argc, char *argv[])
{
    int src, dest;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    if (argc != 3) {
        printf("Usage: %s <source> <destination>\n", argv[0]);
        return 1;
    }

    src = open(argv[1], O_RDONLY);

    if (src == -1) {
        perror("Error opening source file");
        return 1;
    }

    dest = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (dest == -1) {
        perror("Error opening destination file");
        close(src);
        return 1;
    }

    while ((bytes_read = read(src, buffer, BUFFER_SIZE)) > 0) {

        bytes_written = write(dest, buffer, bytes_read);

        if (bytes_written != bytes_read) {
            perror("Error writing file");
            close(src);

```

```
filecopy.c sample.txt
akshi@akshita:~/filecopy_lab$ gcc filecopy.c -o filecopy
akshi@akshita:~/filecopy_lab$ ls
filecopy filecopy.c sample.txt
akshi@akshita:~/filecopy_lab$ ./filecopy sample.txt copy.txt
akshi@akshita:~/filecopy_lab$ ls
copy.txt filecopy filecopy.c sample.txt
akshi@akshita:~/filecopy_lab$ cat copy.txt
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

#define BUFFER_SIZE 4096

int main(int argc, char *argv[])
{
    int src, dest;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    if (argc != 3) {
        printf("Usage: %s <source> <destination>\n", argv[0]);
        return 1;
    }

    // Open source file for reading
    src = open(argv[1], O_RDONLY);
    if (src == -1) {
        perror("Error opening source file");
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    }

    // Read from source and write to destination
    while ((bytes_read = read(src, buffer, BUFFER_SIZE)) > 0) {

        bytes_written = write(dest, buffer, bytes_read);

        if (bytes_written != bytes_read) {
            perror("Error writing to destination file");
            close(src);
            close(dest);
            return 1;
        }
    }

    if (bytes_read == -1) {
        perror("Error reading source file");
    }

    // Close both files
    close(src);
    close(dest);

    return 0;
}
akshi@akshita:~/filecopy_lab$ diff sample.txt copy.txt
akshi@akshita:~/filecopy_lab$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffecf4dd9a8 /* 29 vars */)
= 0
brk(NULL)                               = 0x580dc3e58000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0
x729417de9000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20811, ...}) = 0
mmap(NULL, 20811, PROT_READ, MAP_PRIVATE, 3, 0) = 0x729417de3000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"...

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+++ exited with 0 +++
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akshi@akshita:~/filecopy_lab$ strace --version
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strace -- version 6.19
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close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x734dc44d8000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
akshi@akshita:~/filecopy_lab$ strace -e trace=openat,read,write,close ./file
copy sample.txt copy2.txt
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC)
= 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0\0"...
, 832) = 832
close(3) = 0
openat(AT_FDCWD, "sample.txt", O_RDONLY) = 3
openat(AT_FDCWD, "copy2.txt", O_WRONLY|O_CREAT|O_TRUNC, 0644) = 4
read(3, "#include <stdio.h>\n#include <std"... , 4096) = 1241
write(4, "#include <stdio.h>\n#include <std"... , 1241) = 1241
read(3, "", 4096) = 0
close(3) = 0
close(4) = 0
+++ exited with 0 +++
```